

Q Frog Jump with
k distances.

k

Q $i+1$
 $i+2$
 $i+3$
 $i+4$
.
.
 $i+k$

} allowed to
jump to all
these indices.

rec

$f(ind)$

{

if ($ind == 0$) return 0;

$fs = f(ind-1) + abs$

$(a[ind]$

$- a[ind-1])$

if ($ind > 1$)

$ss = f(ind-2) + abs$

$a[ind-2]$

$- a[ind])$

3rd jump

4th jump ---

return min

}

$i+1$

$i+2$

$i+3$

new recursive

t(ind)

{

if (ind == 0) return 0;
if dp[ind] == -1
min steps = INT - max;

for (j = 1; j <= k; j++)

{ if (ind - j > 0)

jump = t(ind - j)

+ abs(a[ind - j]

min - steps - a[ind])

= min (min steps, jump);

dp[ind] =

return min - steps;

}

T.C $\rightarrow (O(N) \times K)$

S.C $\rightarrow O(N) + O(N)$

↳ Tabulation

```
int dp[n]
```

```
dp[0] = 0
```

```
for i → n
```

```
{
```

```
    min_steps = INT_MAX;
```

```
    for j → k
```

```
    { if (i-j >= 0)
```

```
        s = dp[i-j]
```

```
        + abs(a[i] - a[i-j])
```

```
        min_jump = min(min_steps, j+1)
```

```
    }
```

```
} dp[i] = min_jump;
```

↳ space optimization

↳ $O(N) \rightarrow O(1)$!

↳ $O(N) \rightarrow O(1)$

prev2

dp[i-2]

prev1

dp[i-1]

curr

dp[i]

